IoT Platforms

Frank Walsh

What are IoT platforms

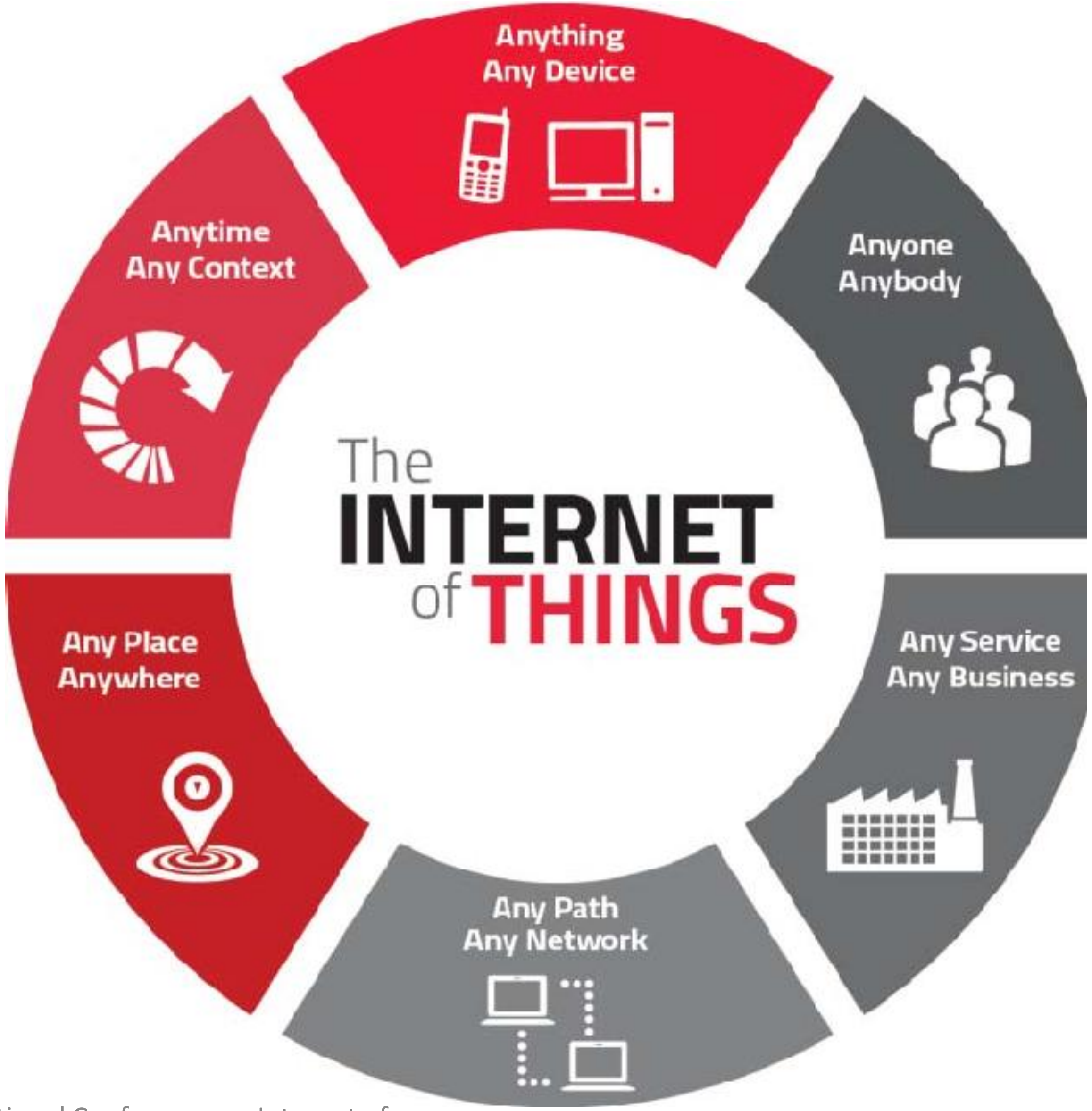
- IoT applications combine sensors, devices, data, analytics and integrations in a seamless and unified way
  - e.g. your project!
- IoT Platforms provide software tools and components to:
  - connect sensors, devices, and data networks
  - Analyse and store data
  - Integrate with other apps
- So what? We know the tech for that now (I2C, SPI, BLE, MQTT, Python...)
- Main selling point of an IoT platform is software that it
  - accelerates the IoT development process
  - Focuses on IoT: brings in best of breed features
  - Provides initial scaffolding for IoT projects

What are IoT Platform

- Many(not all) are cloud-based platforms that require subscription
- Provide device/language agnostic set of Software Development kits
  - Arduino/RPi/beagleboard
- IoT development is generally iterative:
  - Starts with initial simple use case
  - Once operational, data/insights result in new usecased
- IoT platforms should promote scalable, iterative development
  - Allow for quick app development
  - Ability to adapt/optimise apps quickly

IoT Platform Characteristics

- Manage many concurrent device connections
- Connectivity across several connection types
- "Off-the-peg" IoT protocol stack
- Manage/analyse/visualise data
- Integrations to other services/apps
- App Development



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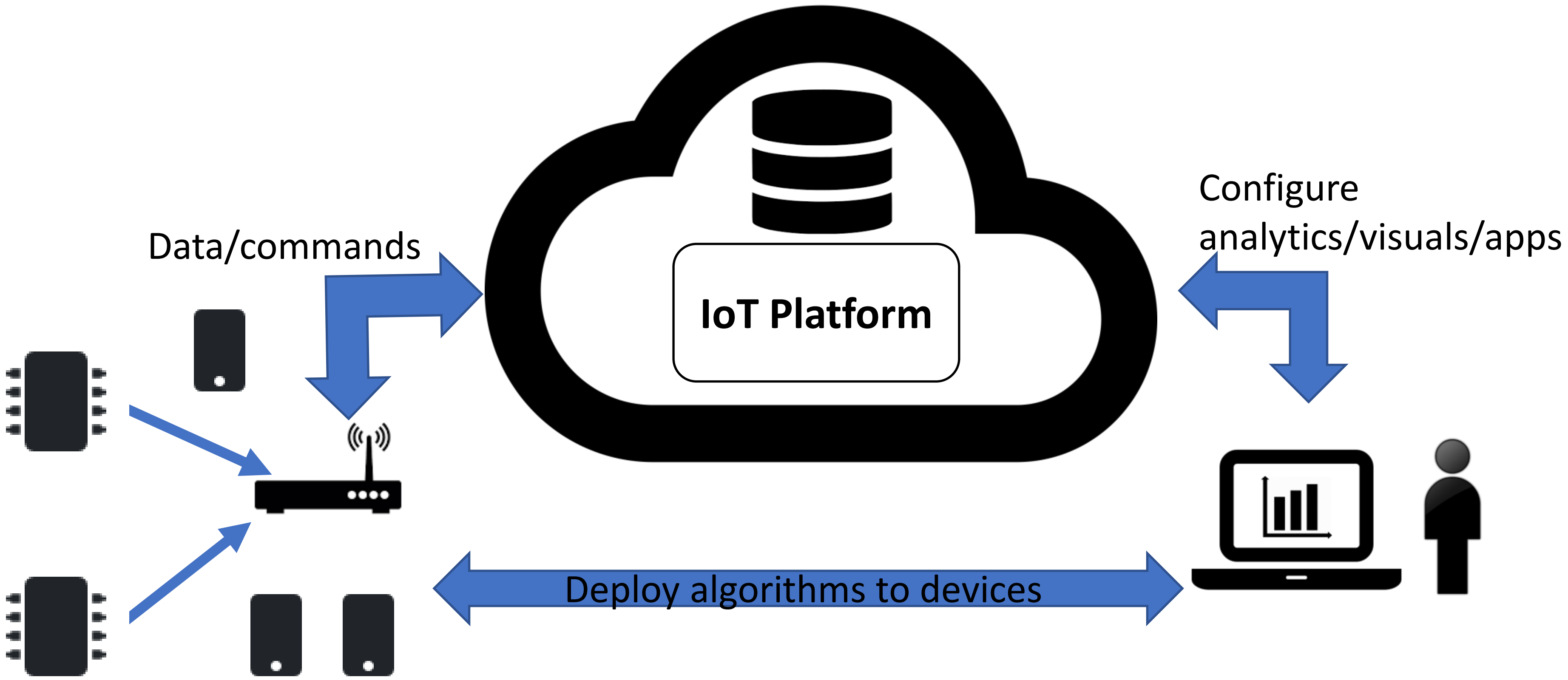
IoT Platform – general architecture



IoT Platform Advantages

- Use sofware component that has been pre-built and pre-tested. This increases the reliability of your application and reduces development effort.
- IoT frameworks constantly evolve, providing new features, integrations etc.
- Encourages better "design pattern" for your IoT app.
- Predefined APIs and docs
    - Great for collaboration
- "Baked-in" standards and features:
    - Security, authentication, scalability...

Which one?

- Connectivity
  - Does the platform provide suitable capability and integrations (WiFi/Cellular/LPWan-Sigfox)
- Maturity
  - In business for long? Critical mass in developer community?
- Free
  - Is there a free tier (handy for evaluation)?
- Service type
  - Platforms try to distinguish themselves – what specialisms/USP does it have?
- Security
  - What security model do they use? Is there security issues reported in past?
- Geographic area
  - Does it operate well at your location (can you select edges/data centres)

Examples:

Mobile Device IoT using Blynk

Frank Walsh

Mobile Devices and IoT

- Critical component of many IoT solutions is the mobile phone/tablet.
  - IOS/Android dominate
- >2.6 billion users worldwide
  - Simple, mobile connection to the internet
- Provides nice features for IoT apps
  - Packed with sensors (Location, accelerometer, camera)
  - Can connect/interlink other smart devices using Bluetooth, BLE, etc.

Mobile Apps in IoT, Examples

- Wearables
  - wristwatches, eyeglasses and rings
- Healthcare
  - Medical sensors obtain health data and transfer to a mobile app.
  - This data can be transferred remotely to doctor/ family members
- SmartHome
  - Nest – see who’s at the door...
- AgriTech
  - MooCall

Low-code IoT cloud platform with user experience at its core

Easily build exceptional, fully customizable mobile and web IoT applications. Securely deploy and manage millions of devices worldwide.

Enterprise Solutions Sign Up Free →

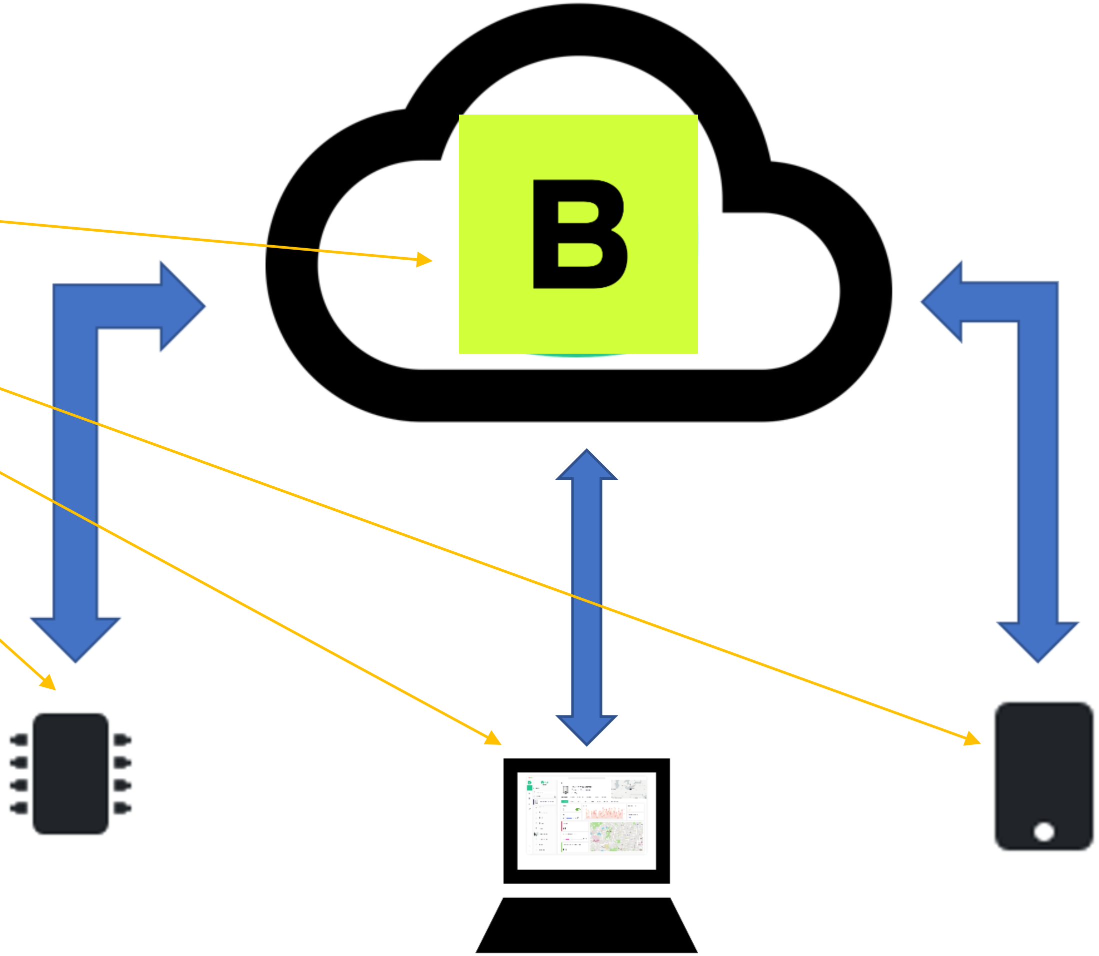
4.6/5 stars

What's Blynk

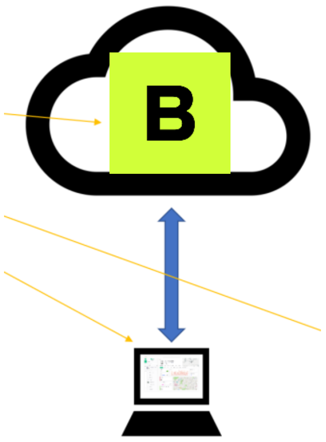
- Yet Another IoT Platform
  - Specialism: mobile application builder
- "Blynk is a full suite of software required to prototype, deploy, and remotely manage connected electronic devices at any scale: from personal IoT projects to millions of commercial connected products."

Platform Overview

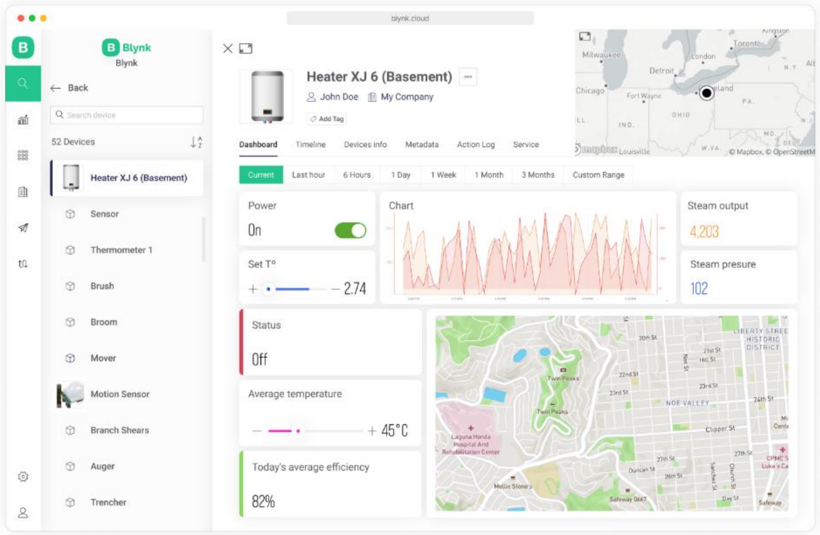
- 4 Compnents
  - Blink.Cloud
  - Blynk.App
  - Blynk.Console
  - Blynk.Edgent (Device Libraries)



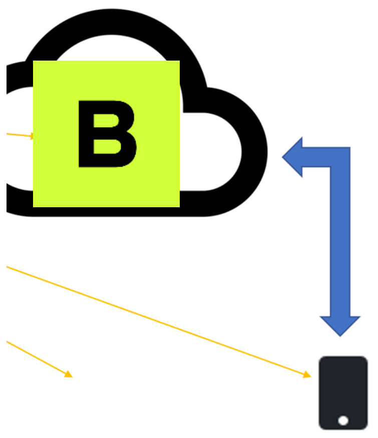
Blynk.Console



- **Blynk.Console** is a feature-rich web application
- Register with Blynk to gain access
- Use it to:
    - Configuration of how connected devices work on the platform + application settings.
    - Management of devices, their data, users, organizations and locations
    - Remote monitoring and control of devices



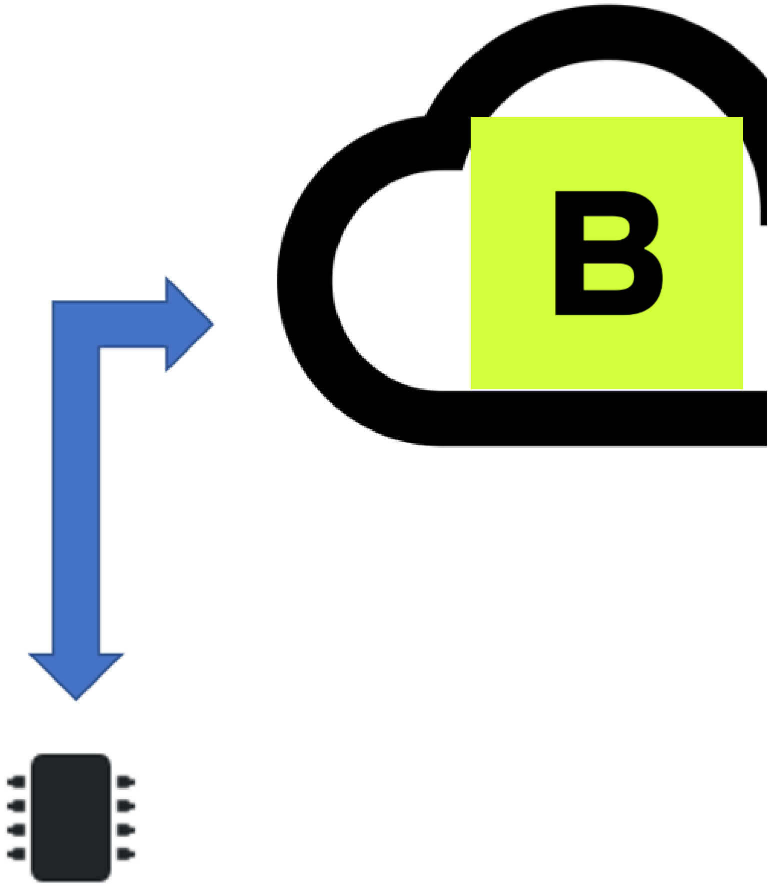
Blynk App



- **Blynk App** is a multi-functional native iOS and Android mobile application
- Use it to:
    - Remote monitoring and control of connected devices that work with Blynk platform
    - Configuration of mobile UI during prototyping and production stages
    - Automate work of connected devices
- Allows you to create mobile applications using drag and drop
    - Uses "widgets"
- No Code!!!
    - Don't worry – that comes later

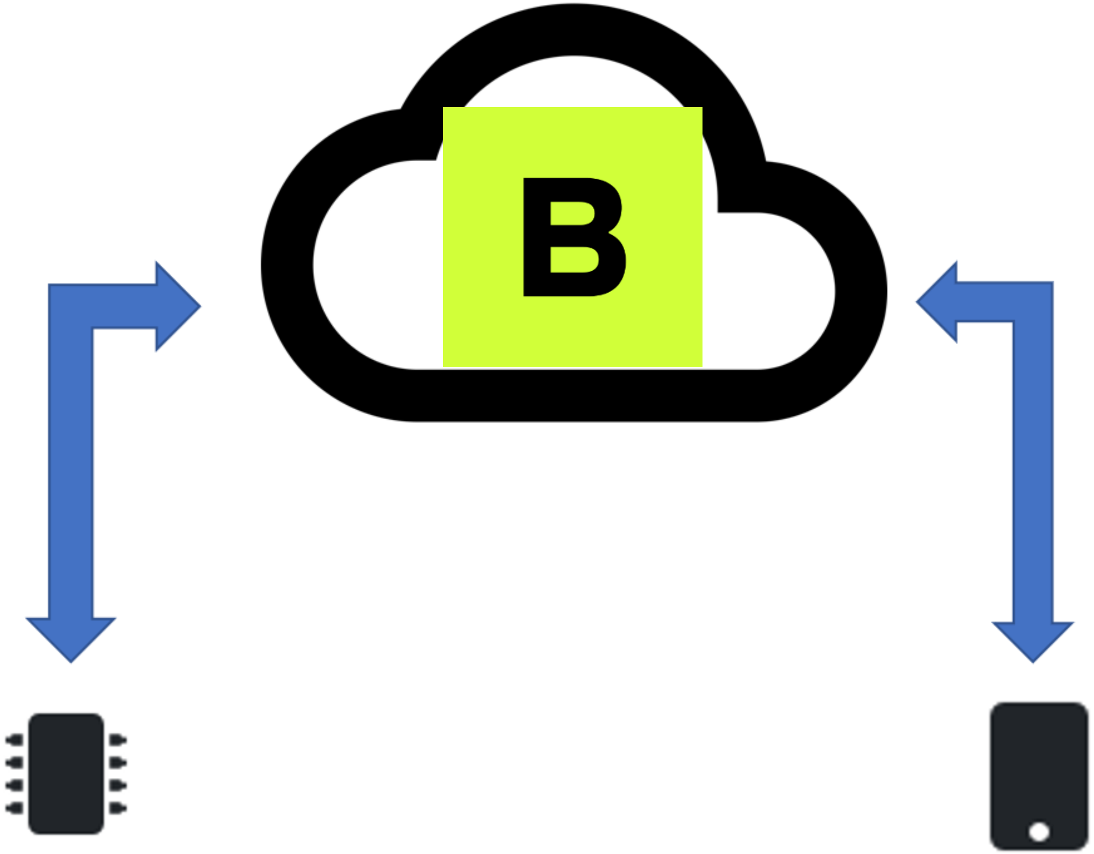
Blynk Library

- Lightweight embedded library that runs on over 400 supported hardware models
    - We’ll use a unsupported python one
- Use it to:
    - bring your device online and authenticating)
    - Connectivity management (WiFi, Cellular, Ethernet)
    - Data transfer Over-the-air firmware updates (for selected hardware models)
- Available for most popular hardware platforms
    - RPi, Arduino, ...
- Enable communication between the device and cloud API to work with specific Blynk App and Blynk cloud features
- Python, Javascript, C++ … on the RPi.



Blynk cloud Platform

- **Blynk Cloud** is a server infrastructure – heart of Blynk IoT platform. Cloud is responsible for binding all the platform components together.
- Controls all the communications between the mobile device(e.g. your phone) and hardware (e.g. the RPi)
- Remember we talked about the benefits of indirect communication.
- It's Cloud-based but you can run your own private Blynk Server
- A bit like MQTT...(actually, after peeking at the source code, it uses the same MQTT libraries from the previous lab!)



Benefits

- Don’t need to be a mobile app developer
- Minimal code
- Reasonably mature (around since 2014)
- Very quick to create a prototype
- If you want, process to publish to App store/Google Play

Similar API & UI for all supported hardware & devices

Connections/ Protocols:

- WiFi
- Bluetooth and BLE
- Ethernet
- USB (Serial)
- GSM
- …

Set of easy-to-use Widgets

Connect device with no code writing

Easy to integrate and add new functionality using virtual pins

History data monitoring via SuperChart widget

Device-to-Device communication using Bridge Widget

Sending emails, tweets, push notifications,

Features

Pick your way to connect to Blynk

Blynk.Edgent (recommended)

Cloud connection, data exchange, WiFi provisioning, OTA firmware updates and more for supported boards and dual-MCU configurations (NCP).

DOCUMENTATION

MQTT API

Ideal for projects utilizing MQTT libraries, Node-RED and similar, or SDKs with MQTT support as well as integrating off-the-shelf MQTT-enabled gateways and hardware.

DOCUMENTATION

HTTPs API

Blynk HTTPs RESTful API allows to easily send and get data from hardware, cloud, and apps using HTTPs requests including sending timestamped data in batches.

DOCUMENTATION

Blynk.NCP

NCP handles Blynk.Cloud connectivity (WiFi, Ethernet, Cellular), offloading this task from the main device's MCU.

Main MCU operates with a lightweight client library, communicating with the

Blynk library

Portable C++ library, pre-configured to work with hundreds of dev boards.

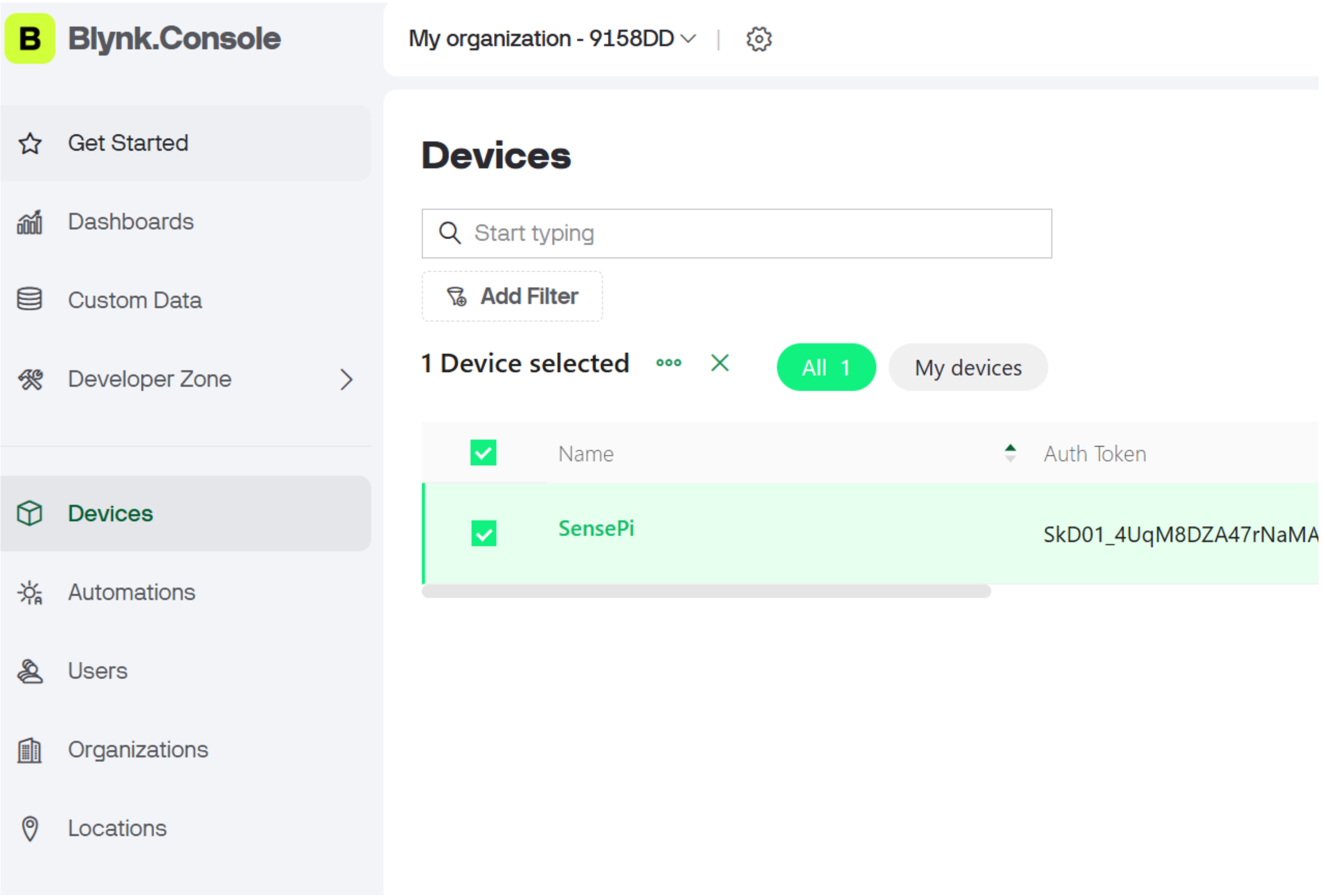
Low latency, bi-directional communication through a streaming connection protocol.

Blynk Concepts

- Devices
- Templates
- Events
- Dashboards

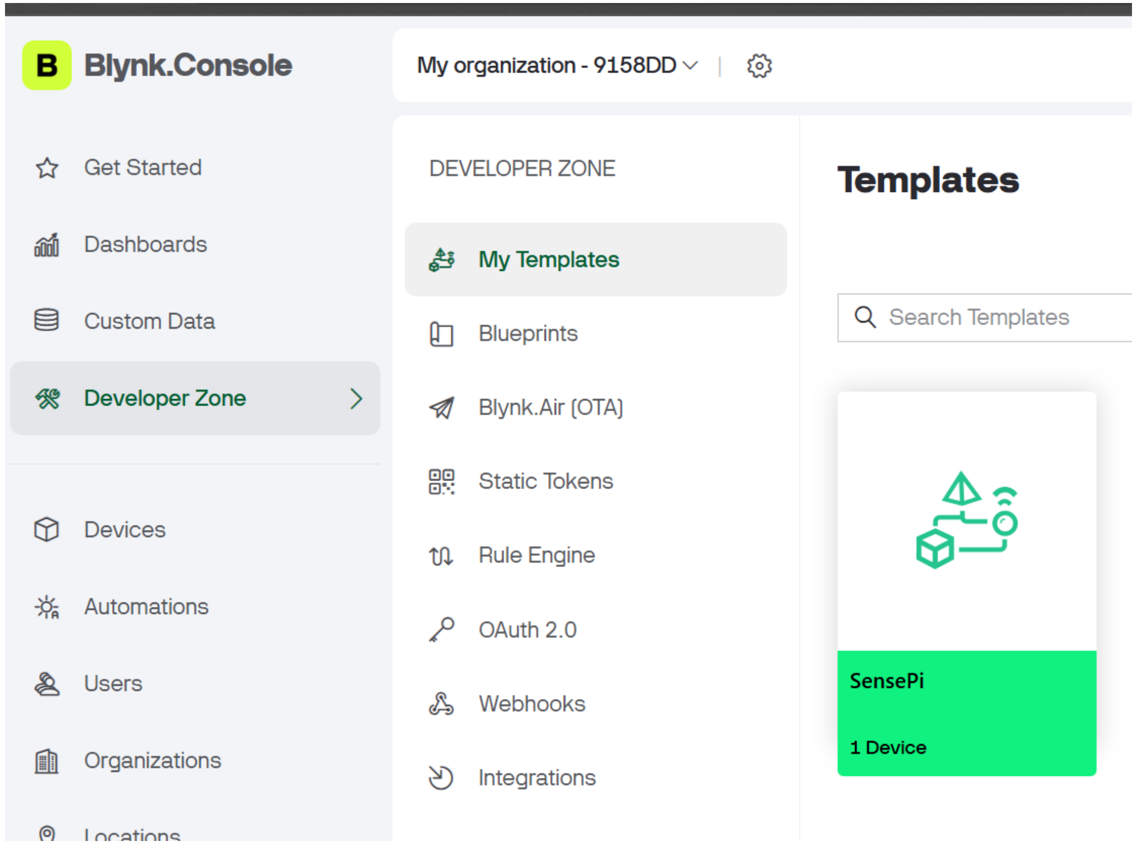
Blynk Concepts: Device

- A "device" can be:
  - A small MicroController based hardware (e.g. Arduino, **Raspberry Pi,** etc)
  - A finished physical product like a Smart Air Conditioner, or a virtual service that sends data to Blynk Cloud using REST API.
- Each device has an "**Authentication Token**". This is a unique identifier generated in Blynk.Cloud
- Every device uses a **Device Template.**



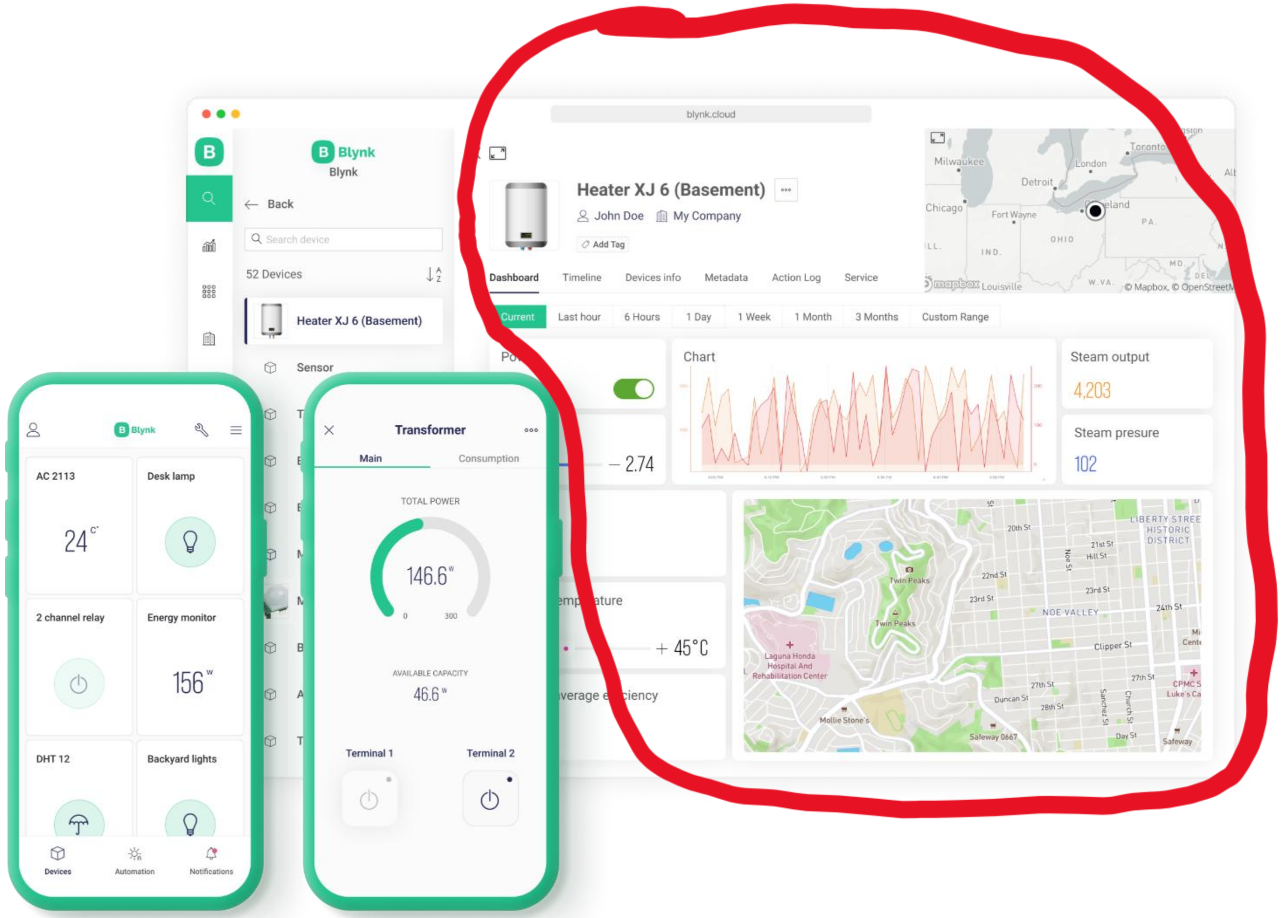
Blynk Concepts: Device Template

- **A Device Template is a set of configurations used by similar devices.**
- A device template specifies:
  - **General Settings:** general settings of the device(name, ID etc..)
  - **Metadata:** Serial Number field, email field to use for email notifications
  - **Datastreams:** channels for any data that flows in and out from the device to the cloud. For example sensor data should go through a Datastream. These are also known as "**Virtual Pins**".
  - **Events:** important events in the life of the device that should be logged and, if needed, used for notifications. Events can be triggered from the device itself or externally using HTTP API
  - **Dashboards:** Set of UI elements to visualise data and send commands/data to/from device
    - There are two dashboards: Web and Mobile



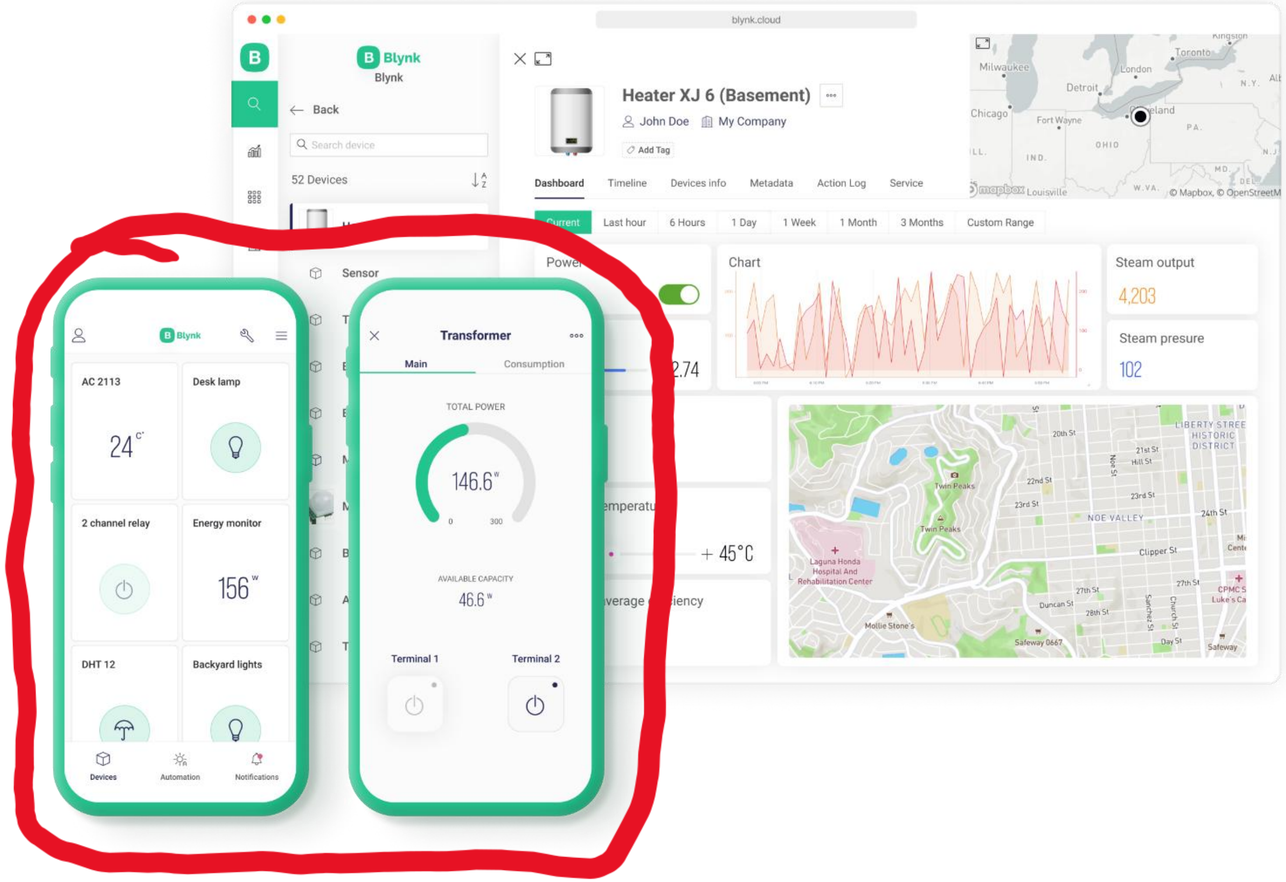
Blynk Concepts: Web Dashboard

- a set of UI elements (widgets) to visualize the data from the device accessible for the users in Blynk.Console – a web-based application.



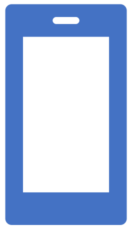
Blynk Concepts: Mobile Dashboard

- a set of UI elements (widgets) to visualize the data in Blynk mobile apps for iOS and Android.
- Mobile apps also contain a template of how device is represented in the list of devices (tiles)



Development Steps

1. Create an Account with Blynk
2. Add a Device
    1. Choose your hardware
    2. Create “Template”
    3. This will result in an Authentication Credentials for that Device
3. Code the Hardware Device(e.g. Rpi)
   Use the relevant library(e.g. Python Library)
4. Install Blynk App on Phone
    1. Develop Blynk

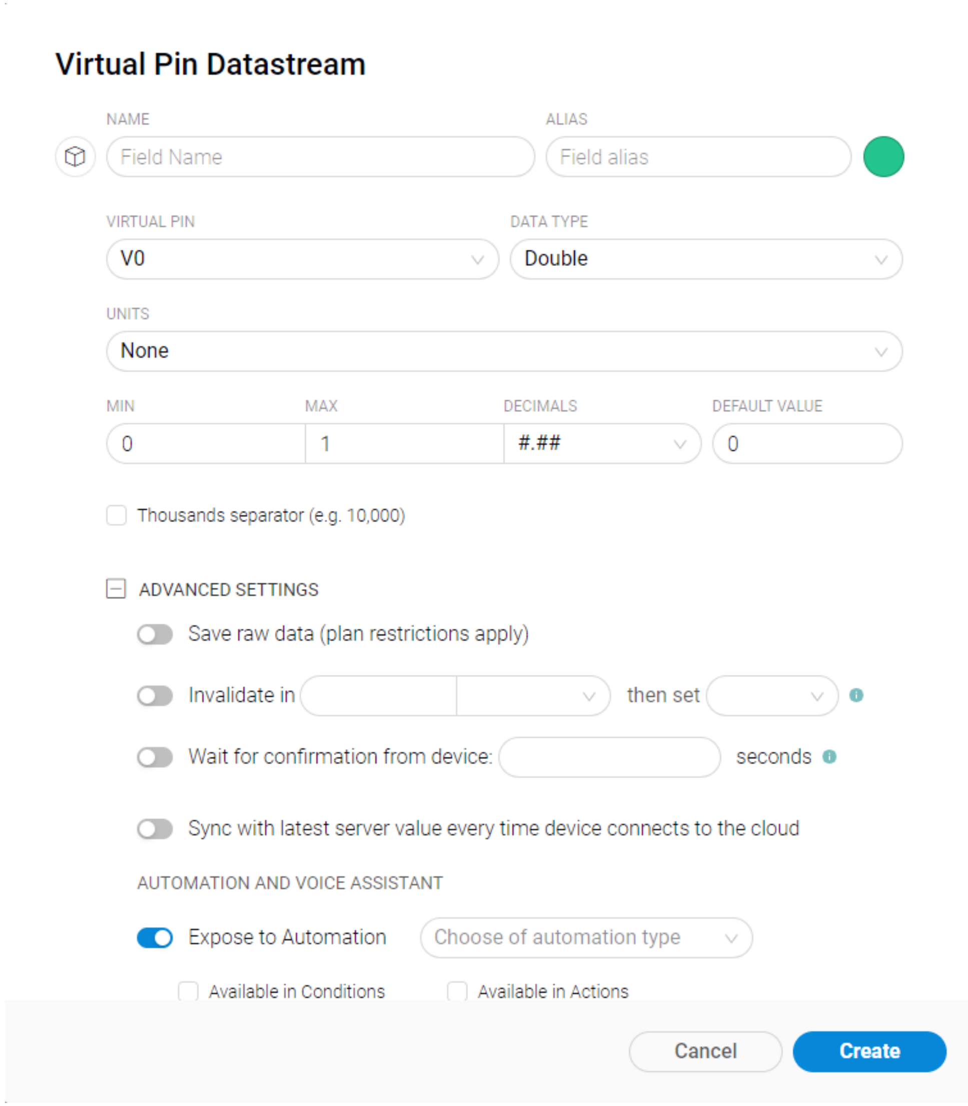


Add a Device

- Create a new device
- Set Notifications:
  - Specify who gets notifications, which events will trigger notifications, and which channel should be used.
  - 3 options:
    - E-mail
    - Push notifications - a push notification is a message that pops up on a mobile device. Users don't have to be in the app to get them
    - SMS - notification will be delivered as a text message using mobile operator (only available in paid option though…)

Working with Templates

- Datastreams is a way to structure data that regularly flows in an out from device.
- Use it for sensor data, any telemetry, or actuators.
- Virtual Pin is a concept invented to provide exchange of any data between hardware, web and mobile app.
- Virtual pins allow you to interface with any sensor, any library, any actuator.Think about Virtual Pins as a box where you can put any value, and everyone who has access to this box can see this value.
- You can set UNITS that will be viewed in the Widget by selecting them from the dropdown menu.
- It's a very powerful feature to display and send any data from your hardware to the application.**Please make sure you differentiate Virtual Pins from physical GPIO pins on your hardware.**

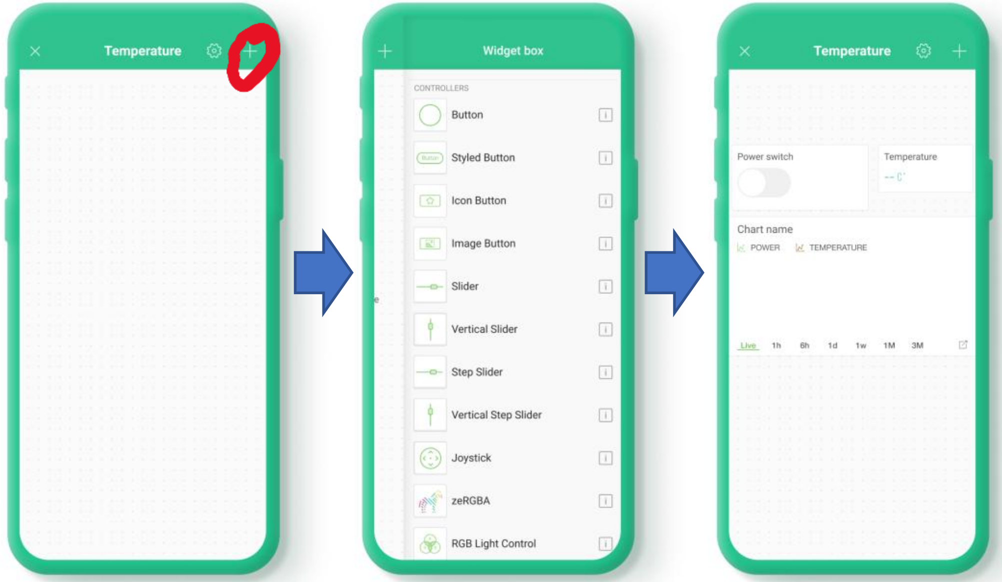


Working with Templates: Events

- Events are used to track important events happening on your devices.
- Events can trigger notifications which can be
  - sent over email
  - delivered as push notifications to user’s phone
  - SMS
- Examples:
  - log a moment when a temperature reached a certain threshold and send a notification to selected users
  - You need to log a total working hours of the device. If it approaches or goes beyond a max value, you would need to notify technical support so that they can replace the device

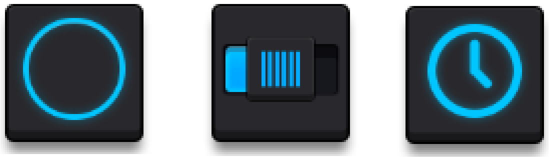
Blynk.Apps

- Using the Blynk App, a Mobile Dashboard can be built from Widgets - modular UI elements which can be positioned on the canvas.
- To add Widgets to Canvas:
  - tap anywhere on empty canvas or press PLUS icon in the top right corner of the app.
  - A Widget Box will open
  - Find the widget you need and tap on it
  - Widget will placed in the first empty area required for this widget
  - Configure Widget.



Widgets

- Categorised as:
  - Controllers



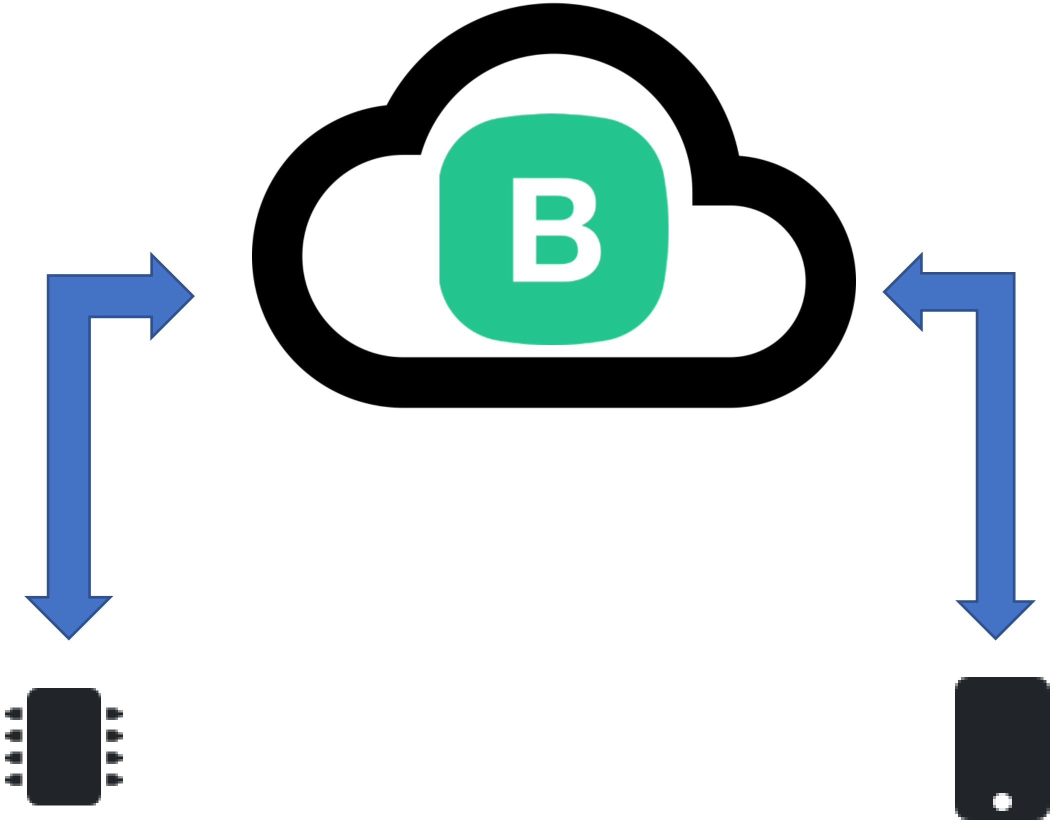
  - Display

  - Notifications

- Interface
  - Tabs, Menu, Map..

Example

DEMO!!!!!



WARNING!!!!

- Blynk has gone very commercial
- Still has “free tier” but just 200,000 messages allowed
- You can burn through that very quickly if you’re not careful

```python
# Main loop to keep the Blynk connection alive and process events
if __name__ == "__main__":
    print("Blynk application started. Listening for events...")
    try:
        while True:
            blynk.run()
            blynk.virtual_write(0, sense.temperature) #ADD THIS LINE!!!
            sleep(2)
    except KeyboardInterrupt:
        print("Blynk application stopped.")
```

- My approach:
    - Just send messages when you need to
    - Stop the app immediately on finishing testing